Facial nerve

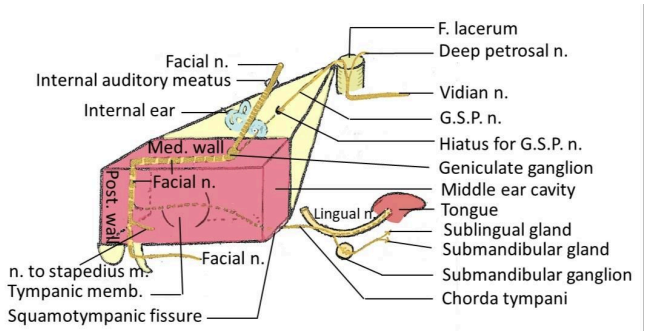
The nerve you protect in every ear and parotid operation: nuclei and fibres, the segments, branches in order, and the surgical landmarks.

1. Components and nuclei
2. Intratemporal segments
3. Branches in order
4. Finding the trunk at parotidectomy
5. Grading facial palsy
6. On imaging
7. In theatre

01 Components and nuclei

- **Motor (special visceral efferent):** muscles of facial expression, stapedius, posterior belly of digastric, stylohyoid (2nd arch).
- **Secretomotor (parasympathetic) from the superior salivatory nucleus:** lacrimal gland and nasal glands (via the greater petrosal nerve and pterygopalatine ganglion); submandibular and sublingual glands (via chorda tympani and submandibular ganglion).
- **Taste:** anterior two-thirds of the tongue (chorda tympani) and the palate (greater petrosal); cell bodies in the **geniculate ganglion**, ending in the nucleus of the tractus solitarius.
- **General sensation:** small area of the concha and posterior canal wall.
- The sensory and parasympathetic fibres travel in the **nervus intermedius**.

> **CLINICAL PEARL** Upper motor neurone lesions spare the forehead because the upper face receives bilateral cortical supply.



Branches of the facial nerve

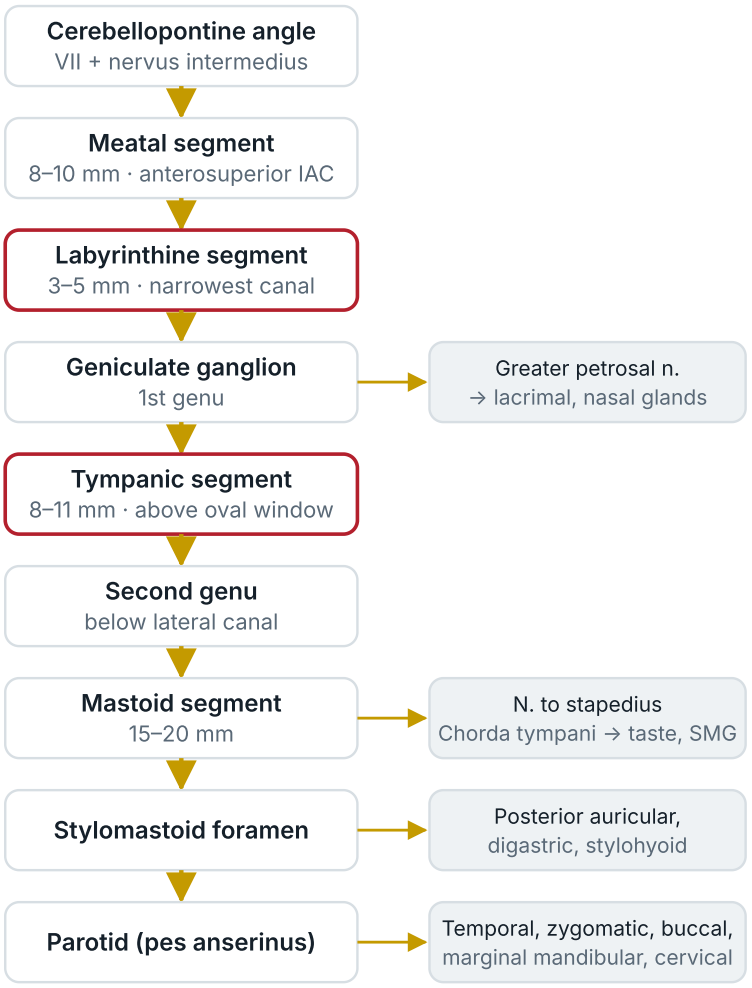
Dr. Sameh Doss handout, p. 149

02 Intratemporal segments

- **Cerebellopontine angle → meatal segment** (8–10 mm) in the anterosuperior IAC above the crista falciformis, in front of Bill's bar.
- **Labyrinthine segment** (3–5 mm): the **narrowest part of the fallopian canal** (about 0.68 mm), from the fundus to the **geniculate ganglion** (first genu), where the **greater petrosal nerve** leaves.
- **Tympanic (horizontal) segment** (8–11 mm): along the medial wall, **above the oval window and below the lateral SCC**, ending near the pyramid at the **second genu**. Most common site of **dehiscence**.
- **Mastoid (vertical) segment** (about 15–20 mm): down behind the posterior wall to the **stylomastoid foramen**; gives the **nerve to stapedius** and **chorda tympani** (usually 3–5 mm above the foramen).

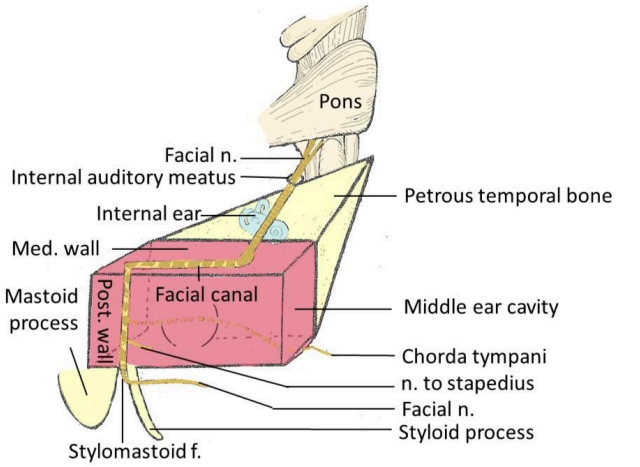
CLINICAL PEARL The labyrinthine segment is narrow and its blood supply is watershed, which is why swelling here is blamed for Bell's palsy.

SOURCES DIFFER MASTOID SEGMENT LENGTH: not given in either handout; most texts give about 15–20 mm (some 13 mm).



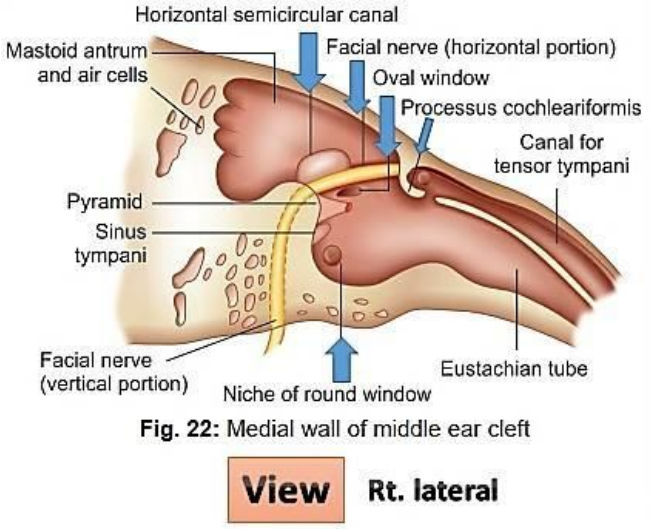
Red outline: narrowest segment and commonest dehiscence.

Facial nerve course: each segment with its length, and the branch that leaves at each level. Lesion level predicts which functions are lost (tears, stapedial reflex, taste).



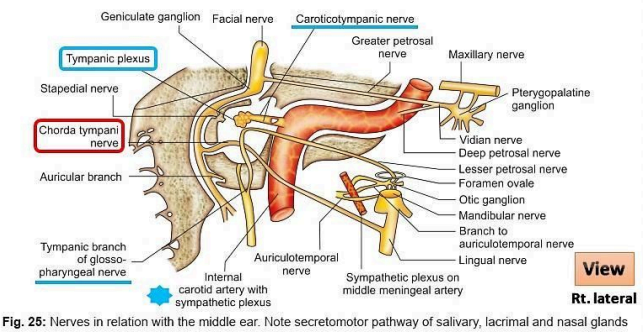
Course of the facial nerve through the temporal bone

Dr. Sameh Doss handout, p. 149



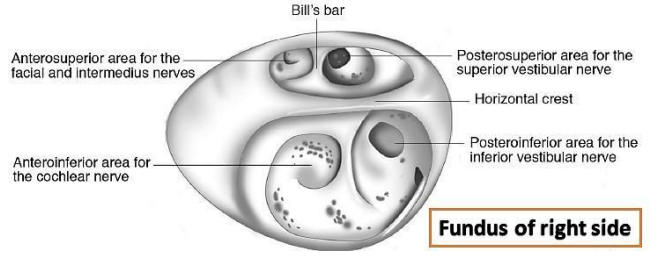
Medial wall of the middle ear cleft

ENT Steps Anatomy, p. 48



Nerves of the middle ear: tympanic plexus and petrosal nerves

ENT Steps Anatomy, p. 53



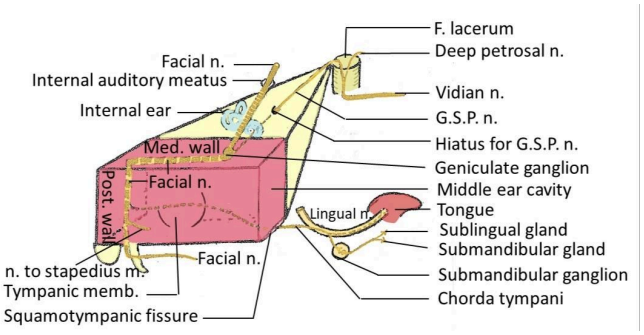
Fundus of the IAC: Bill's bar and the transverse crest

ENT Steps Anatomy, p. 83

03 Branches in order

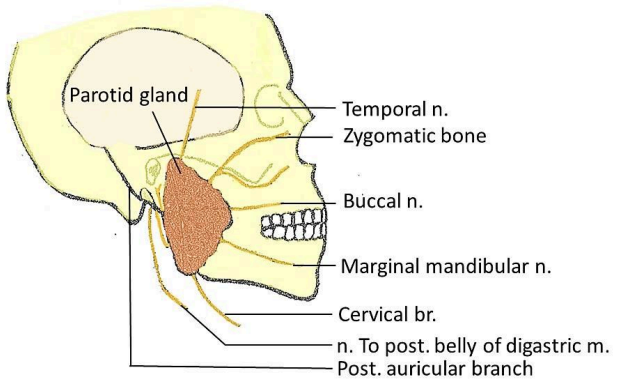
- Inside the temporal bone: **greater petrosal** (geniculate ganglion), **nerve to stapedius**, **chorda tympani**.
- At the stylomastoid foramen: **posterior auricular**, **digastric** (posterior belly) and **stylohyoid** branches.
- In the parotid: divides at the **pes anserinus** into **temporofacial** and **cervicofacial** divisions, giving **temporal, zygomatic, buccal, marginal mandibular and cervical** branches.
- Topognostic tests: Schirmer's (greater petrosal), stapedial reflex (nerve to stapedius), taste and salivary flow (chorda tympani).

> **CLINICAL PEARL** Mnemonic for the facial branches: "To Zanzibar By Motor Car".



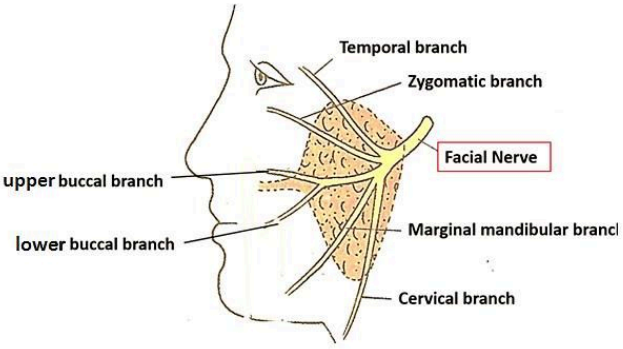
Branches of the facial nerve

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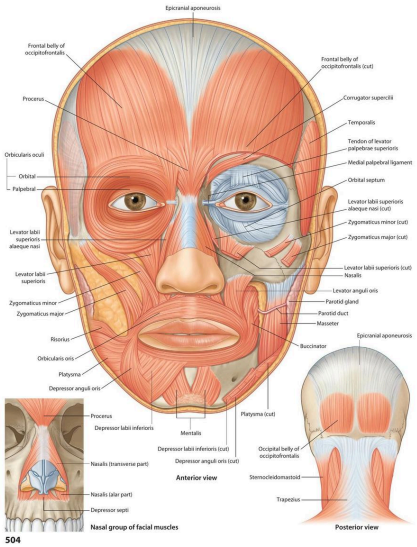
Extracranial branches of the facial nerve

Dr. Sameh Doss handout, p. 149



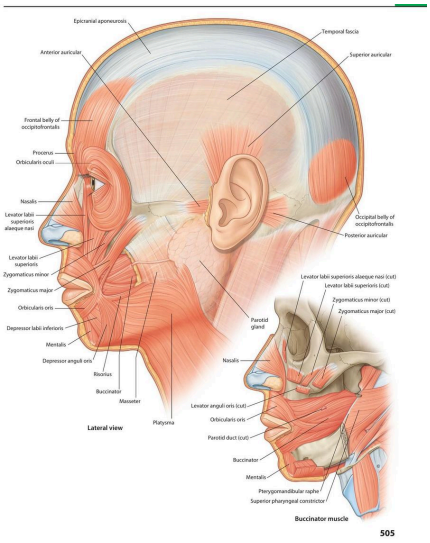
Branches of the facial nerve in the face

ENT Steps Anatomy, p. 183



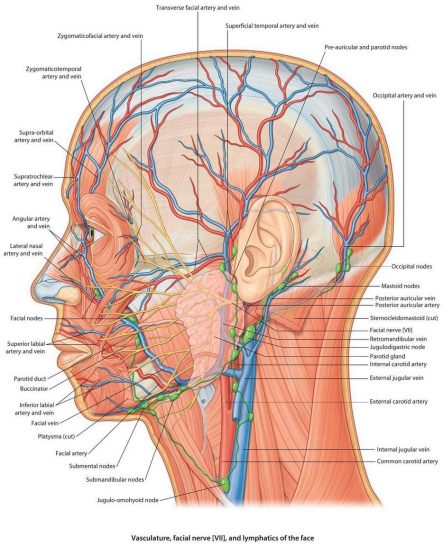
Facial muscles (anterior view), nasal group and posterior view

Gray's Atlas of Anatomy, 2nd ed., p. 504



Facial muscles (lateral view) and buccinator

Gray's Atlas of Anatomy, 2nd ed., p. 505



Vasculature, facial nerve [VII] and lymphatics of the face

Gray's Atlas of Anatomy, 2nd ed., p. 506

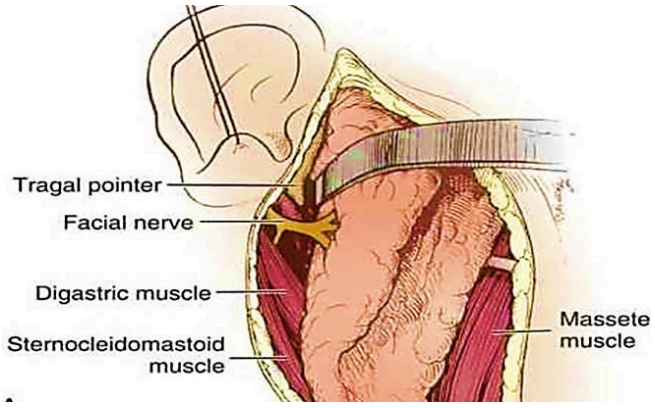
04 Finding the trunk at parotidectomy

- **Tragal pointer:** the nerve lies about **1–1.5 cm deep, medial and inferior** to the tip of the tragal cartilage.

- **Tympanomastoid suture:** the nerve lies **6–8 mm deep** to its outer end (the most reliable bony landmark).
- **Posterior belly of digastric:** the nerve runs just above its upper border near the mastoid attachment, at about the same depth.
- **Styloid process:** the nerve crosses its posterolateral base; it is deeper than the nerve, so it is not a safe landmark to dissect toward.
- **Retrograde approach** from peripheral branches: buccal branch runs about 1 cm above and parallel to Stensen's duct; marginal mandibular branch runs superficial to the facial vessels about 1–2 cm below the mandible.

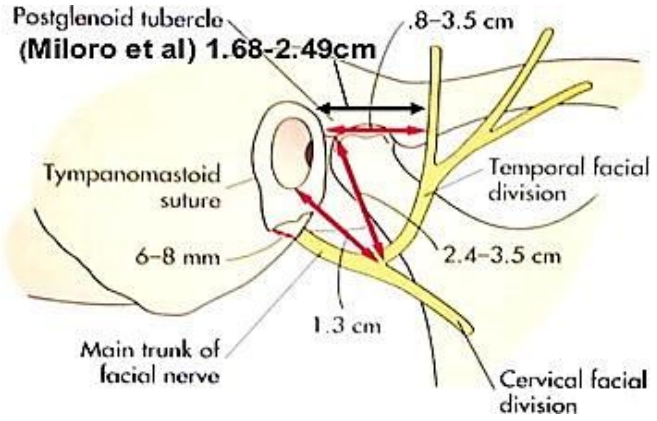
CLINICAL PEARL Identify at least two landmarks before dividing tissue deep to the tragal pointer.

SOURCES DIFFER TRAGAL POINTER: ENT Steps says 1–1.5 cm medial and inferior; Conley's classic figure is 1 cm deep, anterior and inferior.



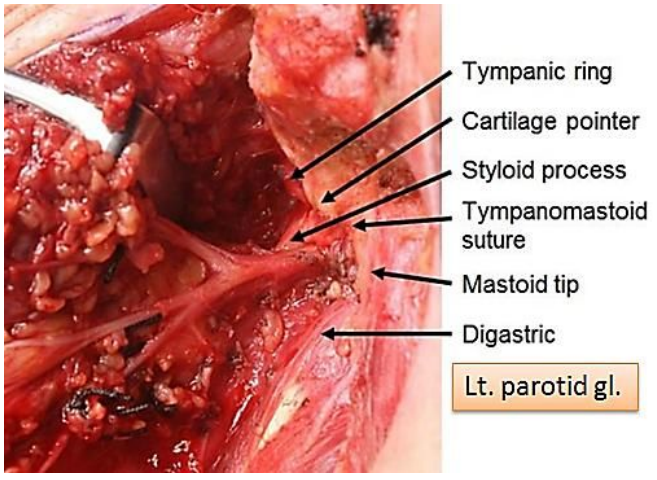
Landmarks for the facial nerve trunk: tragal pointer, digastric, mastoid

ENT Steps Anatomy, p. 185



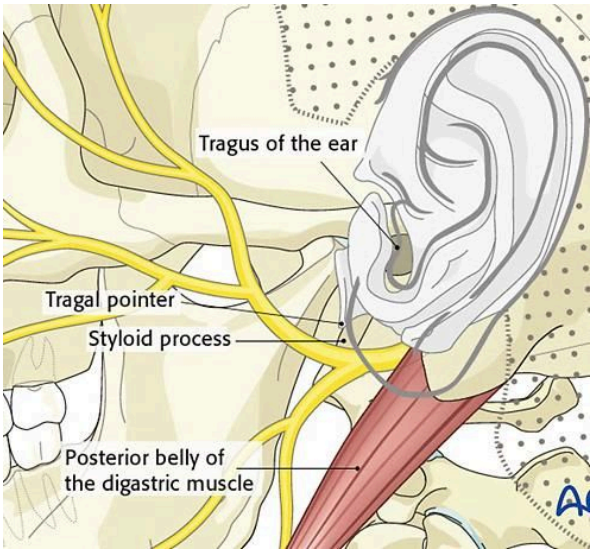
Distances from landmarks to the facial nerve trunk

ENT Steps Anatomy, p. 186



Surgical landmarks of the facial nerve at parotidectomy

ENT Steps Anatomy, p. 187



Tragal pointer, styloid process and posterior belly of digastric

ENT Steps Anatomy, p. 186

05 Grading facial palsy

- **House–Brackmann:** I normal; II mild (slight weakness, normal at rest); III moderate (obvious but not disfiguring, eye closes with effort); IV moderately severe (incomplete eye closure); V severe (barely

perceptible movement); VI total paralysis.
- Grade IV and above is defined by incomplete eye closure, which needs eye protection.

06 Radiological anatomy: On imaging

- **HRCT:** bony canal, dehiscence, fractures (longitudinal fractures injure the nerve near the geniculate ganglion; transverse fractures more often cause palsy).
- **MRI with gadolinium:** enhancement of the labyrinthine segment and geniculate ganglion in Bell's palsy and Ramsay Hunt; schwannomas, haemangiomas and perineural spread of parotid cancer.
- Imaging is needed for palsy that is gradual, recurrent, painful, with other cranial nerve signs, or not recovering after 3–6 months.

07 Surgical anatomy: In theatre

- **Intraoperative monitoring** with EMG electrodes in orbicularis oculi and oris is routine in otology and parotid surgery; it does not replace anatomical knowledge.
- **Decompression** for trauma or Bell's palsy targets the labyrinthine segment and meatal foramen (middle fossa approach) when electroneurography shows over 90% degeneration within 2 weeks.
- **Repair:** direct tension-free neurorrhaphy, or a great auricular or sural nerve cable graft; hypoglossal–facial or masseteric–facial transfer when the proximal stump is unavailable.

T Tables

Facial nerve segments

Segment	Length	Key points
Meatal (IAC)	8–10 mm	Anterosuperior quadrant, in front of Bill's bar
Labyrinthine	3–5 mm	Narrowest (~0.68 mm); ends at the geniculate ganglion → greater petrosal nerve
Tympanic (horizontal)	8–11 mm	Above oval window, below lateral canal; commonest dehiscence
Mastoid (vertical)	15–20 mm	Nerve to stapedius, chorda tympani; exits at stylomastoid foramen
Extratemporal	—	Posterior auricular, digastric, stylohyoid; then 5 branches in the parotid

House–Brackmann grading

Grade	Description	Eye closure	Mouth
I	Normal	Normal	Normal
II	Mild: slight weakness on close look	Complete with minimal effort	Slight asymmetry
III	Moderate: obvious, not disfiguring; synkinesis may be present	Complete with effort	Slightly weak with maximum effort
IV	Moderately severe: disfiguring asymmetry	Incomplete	Asymmetric with maximum effort
V	Severe: barely perceptible movement	Incomplete	Slight movement
VI	Total paralysis	None	None

Incomplete eye closure starts at grade IV.

M Mnemonics

Facial nerve branches

To Zanzibar By Motor Car

Temporal, Zygomatic, Buccal, Marginal mandibular, Cervical

Intratemporal branches in order

GSC

Greater petrosal, nerve to Stapedius, Chorda tympani

Segments

My Lovely Tiny Mastoid

Meatal, Labyrinthine, Tympanic, Mastoid

Branches inside bone

Greater, Stapes, Chorda (top to bottom)

Greater petrosal, nerve to stapedius, chorda tympani

Q Rapid recall

1. **Narrowest segment of the fallopian canal?** Labyrinthine segment (~0.68 mm).
2. **Which segment of the facial nerve is most often dehiscent?** Tympanic (horizontal) segment above the oval window.
3. **Branches of the facial nerve inside the temporal bone?** Greater petrosal, nerve to stapedius, chorda tympani.
4. **Depth of the facial nerve trunk below the tympanomastoid suture?** 6–8 mm.
5. **Where does the nerve lie relative to the tragal pointer?** About 1–1.5 cm deep, medial and inferior to its tip.
6. **Which House–Brackmann grade first has incomplete eye closure?** Grade IV.
7. **Five terminal branches of the facial nerve?** Temporal, zygomatic, buccal, marginal mandibular, cervical.
8. **Where are the cell bodies of taste fibres in VII?** Geniculate ganglion.
9. **Which nucleus gives VII's parasympathetic fibres?** Superior salivatory nucleus.
10. **Where does the greater petrosal nerve leave VII?** At the geniculate ganglion.
11. **Where does the chorda tympani usually leave the facial nerve?** Mastoid segment, about 3–5 mm above the stylomastoid foramen.
12. **What is the pes anserinus?** The division of the facial nerve in the parotid into temporofacial and cervicofacial trunks.
13. **Why is the forehead spared in a UMN facial palsy?** The upper face receives bilateral cortical supply.
14. **What does Schirmer's test assess?** Lacrimation via the greater petrosal nerve.
15. **Electroneurography threshold for decompression?** Over 90% degeneration within 2 weeks of onset.
16. **Facial nerve in the newborn: where is it vulnerable?** At the stylomastoid foramen, which is superficial because there is no mastoid process.
17. **Which branch of VII supplies the posterior belly of digastric?** Digastric branch, given just after the stylomastoid foramen.
18. **Facial palsy with a dry eye localises the lesion where?** At or proximal to the geniculate ganglion (loss of the greater petrosal nerve).
19. **Facial palsy with hyperacusis but normal tearing localises where?** Between the geniculate ganglion and the nerve to stapedius (tympanic segment).

Figures from ENT Steps Anatomy (based on Bansal) and Dr. Sameh Doss's Anatomy of Head & Neck handout, for personal study.